

Activity 4 (Alternative) – Unlocking the Impossible

Prompt-Chaining for Complex Teaching & Work Tasks

Purpose

Tackle a task that currently feels **too big or too hard** by breaking it into a simple **3-step AI-supported workflow**:

1. **Clarify** the problem and context
2. **Generate** options or a first draft
3. **Critique and refine** the output

You stay in charge as the human **orchestrator** – AI is used as a set of small assistants, not a fully automated “agent”.

This activity pairs well with the companion resource on the website: “**Prompt-Chaining Techniques**”.

Core Activity (Teaching / Pedagogy)

Step 1 – Choose a daunting task

Pick one task in your teaching that feels complex, time-consuming, or “impossible” to get started on. For example:

- Designing a new tutorial that makes student thinking visible about a tricky concept
- Planning a new online module or short course
- Creating a bank of feedback comments for a common assignment
- Sketching an AI-aware, authentic assessment idea

Write it here in one sentence:

My daunting task:

Step 2 – Design a 3-step prompt chain

On paper, design **three prompts**, each with a different job:

1. **Clarify the problem / context**
2. **Generate options or a first draft**
3. **Critique and refine** the output

You can use the templates below and adapt them.

Step 1 – Clarify

You are a **learning designer**.

CONTEXT: I teach [**year level, discipline**]. My students struggle with [**concept / skill**].

I want help with this task: [**describe your daunting task**].

Ask me **5 clarifying questions** that will help you support me effectively.

Answer 2–3 of the questions in your own words.

Step 2 – Generate

Using my answers, propose **2–3 options** to help with this task: [**repeat your task**].

For each option, include:

- A short description
- What I would need to do
- How it helps my students' learning / my workload
- Any obvious limitations

Step 3 – Critique and refine

Now **critique your own options**.

- Which option seems most realistic for my context, and why?
- Suggest **3 small modifications** that would make it more practical for my time and constraints.

Then provide a short, numbered plan (3–5 steps) for how I could pilot this in my next teaching period.

Rewrite these prompts in your own words to match your task.

Step 3 – (Optional) Run part of your chain

If you have AI access and time:

1. Run your **clarify** prompt and answer 1–3 of the questions.
2. Run your **generate** prompt.
3. Skim the output and run your **critique/refine** prompt if time allows.

Ask yourself:

- Which step (clarify, generate, critique) added the **most value** for you?
- What would you still need to change as the human in the loop?

If you don't have AI access, simply **design the chain on paper** – it becomes a playbook you can try later.

Example: Designing a Tutorial Activity (for reference)

Here is a fully worked example you can adapt.

Daunting task: “Design a 20-minute tutorial that makes student reasoning about [concept] visible, not just their final answers.”

1. Clarify prompt:

You are a learning designer.

CONTEXT: I teach a [year level] [discipline] unit and students struggle with [concept].

I want a 20-minute tutorial activity that reveals their reasoning, not just final answers.

Ask me **5 questions** to understand my students and constraints.

2. Generate prompt:

Using my answers, propose **2 different 20-minute activities** that help students surface their reasoning about [concept].

For each activity, include: what students do, how I run it, and how their thinking becomes visible (not just final output).

3. Critique/refine prompt:

Critique your two activities:

- Which one is more likely to reveal student reasoning, and why?
- Suggest **3 changes** to make it realistic for my class size and room setup.
- Give me a short, step-by-step script I could follow in class.

Variants for Researchers and Professional Staff

You can use the same 3-step chaining pattern for other complex tasks.

If you are mainly doing research

Choose a daunting research task, such as:

- Planning a new study or data collection strategy
- Structuring a literature review into themes and gaps
- Drafting a grant “impact” or “significance” section

Design a 3-step chain:

1. Clarify: Ask AI to interview you about your project (5–7 questions).
2. Generate: Ask it to suggest 2–3 possible structures/plans based on your answers.
3. Critique/refine: Ask it to critique its own suggestions and propose improvements or a step-by-step action plan.

If you are in a professional / administrative role

Choose a complex work task, such as:

- Redesigning an information session or onboarding process
- Creating a series of student communications across a semester
- Improving a recurring workflow in your team

Design a 3-step chain:

1. Clarify: Ask AI to generate clarifying questions about your audience, constraints, and goals.
 2. Generate: Ask for 2–3 options (e.g., session outlines, comms calendars, process maps).
 3. Critique/refine: Ask it to critique its options, identify risks, and suggest small, realistic steps you could trial.
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Reflection / Notes

Use the prompts below and the space provided.

- What task did you choose as your “impossible” or daunting problem?
- Which step of the chain (clarify, generate, critique) felt most powerful for you?
- How might you use a similar 3-step workflow in your teaching, research, or professional role?

(Notes:)

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